Innovative Evaluation Practices for BCSE308P - Computer Networks Lab

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FAULT-TOLERANT ROUTING: PERFORMANCE COMPARISON OF RIP AND OSPF CONVERGENCE UNDER NETWORK FAILURES

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ABSTRACT

This project compares the fault-tolerance behaviour of RIPv2 and single-area OSPF on a four-router ring in Cisco Packet Tracer. Four Cisco 2911 routers connect four switched LANs. The study presents recorded link failure, router failure and restoration observations through an offline evidence-replay dashboard.

Observed Connectivity Recovery Time (OCRT) is the external-stopwatch interval from failure introduction until ping replies become consistently successful. For the recorded R1–R2 failure, RIP required approximately 26 seconds and OSPF approximately 7 seconds, each from a single run. Routes changed from the direct connection to the longer ring path. Three other link timings are Assumed (estimated – not measured), retained as illustrative inputs. Across the four supplied links, including those three assumptions, descriptive arithmetic averages are 25.75 seconds for RIP and 7.00 seconds for OSPF. The reduction of these averages is 72.82%; it is not a repeated-trial or statistically significant result.

For R4 failure, RIP's later run recorded approximately 11 seconds, while OSPF's earlier run showed no loss and no measurable OCRT. These runs are not directly comparable. Both protocols returned to their preferred route after link restoration. OSPF's approximately 11-second restoration interval indicates a TTL/path change rather than outage recovery.

Saved configurations were inspected without altering the original files. The dashboard distinguishes captured behaviour, assumed estimates and illustrative paths. Stopwatch reaction, simulator processing, end-to-end effects, single runs and missing loss summaries limit the conclusions. Repeated matched trials and automated timestamping form the next steps.

Certificate

This is to certify that the project report titled "Fault-Tolerant Routing: Performance Comparison of RIP and OSPF Convergence Under Network Failures" is a bonafide record of the work done by Harshith Reddy V (24BCE2479) and Sinchana B Kurkuri (24BDS0352) in partial fulfilment of the requirements of the Innovative Evaluation Practices for BCSE308P - Computer Networks Lab during the Fall Semester 2026-2027.

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1 INTRODUCTION

1.1 Background and motivation

A routed network needs an alternate path when a link or transit router becomes unavailable. Redundant cabling supplies that path, but the routing process must also recognise the changed topology and install usable forwarding information. Temporary interruption can therefore occur even when a physical alternate route exists. This project studies that behaviour in a controlled educational setting, using RIP as the baseline and OSPF as the comparison protocol [1], [2].

1.2 Problem statement

The practical question is how recorded connectivity and route selection change after a failure on an identical four-router topology. The study compares a directly used R1–R2 link failure, a recorded R4 router failure and restoration of the original link. The comparison must retain the distinction between real timing observations, supplied estimates and missing measurements. It must also avoid treating a browser animation as evidence of a network experiment.

1.3 Objectives

The objectives are to verify the common topology and addressing, inspect the saved routing configurations, compare recorded link-failure OCRT, document alternate and restored routes, and provide an offline dashboard whose values can be traced to source tables and screenshots. A further objective is to make the methodology and limitations easy to explain during the lab demonstration and viva.

1.4 Scope and limitations

The scope is four Cisco 2911 routers, four Cisco 2960 switches and four PC-PT devices in a plain ring. Only RIPv2 and single-area OSPF are compared. Existing Packet Tracer experiments are preserved; no new experiment is claimed. The R1–R2 timings are single-run observations. The other three link timings are Assumed (estimated – not measured). No completed ping summaries or repeated-trial data support loss percentages, trial means, standard deviations or confidence intervals.

1.5 Mapping to course modules and the lab experiment

Course topic	Project application
Network layer and routing	RIP distance-vector and OSPF link-state route selection
IP addressing and subnetting	/30 transit links and /24 LANs with static PC gateways
Fault tolerance	Alternate paths following link or transit-router failure
Network simulation	Recorded Packet Tracer topology and CLI/ping observations
Performance interpretation	OCRT, protocol-specific route metrics and provenance-labelled analysis

2 LITERATURE SURVEY

2.1 Overview of the existing protocols

RIPv2 extends RIP with subnet information and supports classless routing. Its distance-vector mechanism advertises destination distances, with hop count as the metric and a bounded reachable distance [1], [3]. OSPFv2 maintains link-state information and computes routes using a shortest-path calculation. Its route metric is cumulative cost rather than RIP hop count [2]. These are established protocols; the project implements no new routing algorithm.

Cisco documents timer and configuration behaviour for RIP and OSPF [4], [5]. These references explain theory and defaults. A configured default is not itself a measured recovery time, and neither protocol specification predicts the stopwatch result for a particular Packet Tracer run.

2.2 Review of related work

Table 2.1 reviews verified primary standards and vendor documentation. Standards establish protocol mechanisms, while command references establish implementation terminology. They are not repeated-run studies of this project topology.

S. No	Authors, year	Technique	Metrics and tool	Merits	Gaps
1	Malkin, 1998 [1]	RIPv2 specification	Hop-count routing; protocol specification	Defines classless RIP behaviour	No Packet Tracer OCRT benchmark
2	Moy, 1998 [2]	OSPFv2 link state and SPF	Cost, LSDB and adjacency procedures	Defines route computation and failure processing	No stopwatch results for this topology
3	Hedrick, 1988 [3]	Original RIP distance vector	Distance and routing updates	Explains RIP foundations	Predates RIPv2 extensions

S. No	Authors, year	Technique	Metrics and tool	Merits	Gaps
4	Cisco [4]	IOS RIP commands	Configuration and timer parameters	Maps theory to IOS settings	Defaults do not isolate observed recovery
5	Cisco [5]	IOS XE OSPF configuration	Hello/Dead and area configuration	Explains operational settings	Vendor documentation is not this experiment

Table 2.1. Literature survey using primary, verifiable sources. The project makes no claim to outperform existing literature.

2.3 Research gap and justification of novelty

The educational gap addressed here is the connection between a routing mechanism, its visible route changes and an honestly bounded end-to-end observation. The contribution is a reproducible evidence organisation and comparison on one topology, a clearly defined OCRT metric, and an offline replay dashboard with measured/assumed status. It is a controlled lab comparison rather than an original protocol or a general performance ranking.

3 PROPOSED SYSTEM

3.1 System architecture and topology

The four routers form R1–R2–R3–R4–R1. Each router connects through its own switch to one PC. Figure 3.1 shows the original topology. The ring retains a path between surviving routers after one ring link or one transit router fails. A failed router's own LAN is not protected by a second gateway in this design.

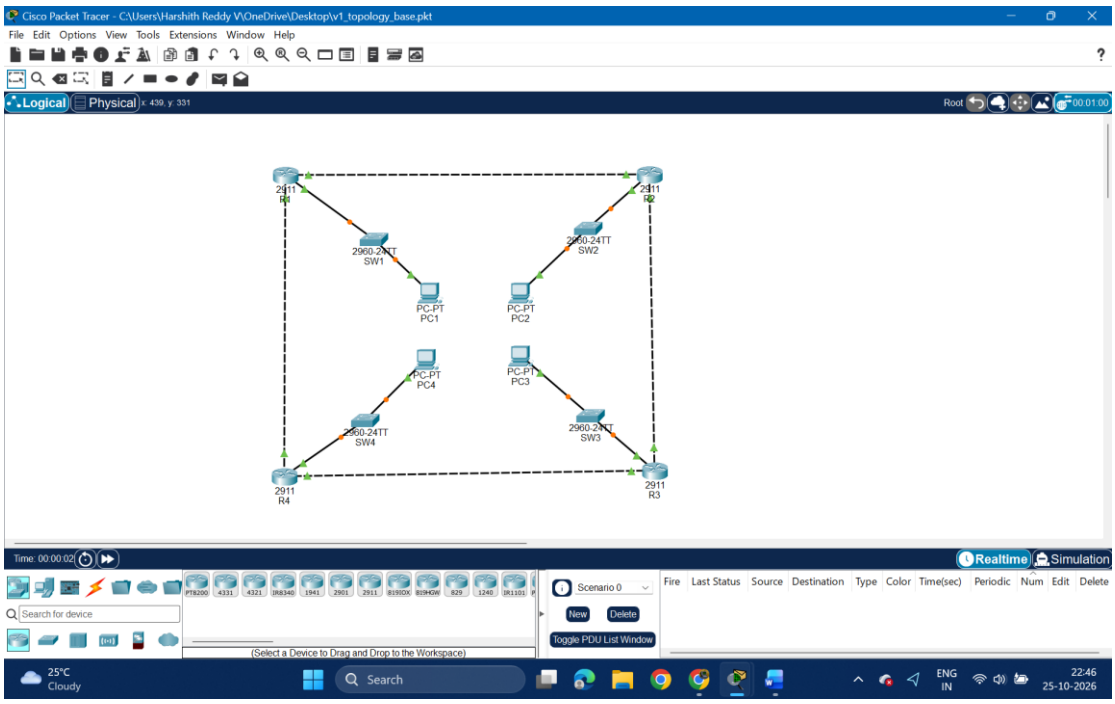


Fig. 3.1. Original ring topology with four routers, switches and PCs. Recorded evidence (Packet Tracer), E06.

3.2 Baseline mechanism description

The baseline is RIPv2. The saved RIP file contains router rip, version 2, network 10.0.0.0, the local LAN network, and no auto-summary on all four routers. RIP route output identifies learned routes with R and administrative distance 120. The R1 route to PC2's LAN initially has hop metric 1; after R1–R2 fails it uses R4 with hop metric 3.

3.3 Experimental procedure and pseudocode

The comparison mechanism is OSPF process 1 in area 0 on the same saved addressing. The procedure below describes the recorded experiment and its analysis; it is not a new routing algorithm. The three assumed links enter only the analysis stage, without a claimed experimental run.

```
FOR each recorded protocol/scenario:
  verify the before route; start continuous PC ping
  trigger failure; start external stopwatch
  after consecutive successful replies, record OCRT
  preserve ping/route evidence; restore the component
END
label supplied estimates separately, with no timed trial
derive labelled arithmetic; validate evidence and replay
```

3.4 Novelty statement

The work contributes a side-by-side RIPv2/OSPF evidence presentation, an explicitly limited end-to-end timing definition, and an evidence-linked offline interface. The browser changes selected scenarios and illustrative paths; it does not execute a routing protocol. The underlying Packet Tracer experiments and original binary files remain unchanged.

3.5 Assumptions and design decisions

Three additional link times are retained as Assumed (estimated – not measured), based on the team's observations of R1–R2 and expected protocol behaviour. Each assumption has zero timed trials and no associated experiment screenshot. The four-case arithmetic average requested by the team includes these estimates, so its interpretation is descriptive and illustrative. It cannot replace repeated timing of the links. Missing packet summaries are not estimated.

4 IMPLEMENTATION

4.1 Tools, platform, versions and hardware

Item	Verified or supplied value
Packet Tracer saved-file version	9.0.1.0858, present in all three decoded file headers
Test machine (user-designated current system)	Intel Core Ultra 5 125H; 16,568,156,160 bytes RAM (approximately 16 GB installed class)
Operating system	Microsoft Windows 11 Home Single Language, 64-bit, version 10.0.26300
Frontend	HTML, CSS, vanilla JavaScript; Chart.js 4.4.8 bundled locally
Offline analysis	Python 3.12; pinned NumPy, pandas, matplotlib and seaborn in Scripts/requirements.txt
Faculty	Dr. Sreethar S, Associate Professor Grade 1

The listed system is the designated test platform. Hardware values were read from that system. Read-only decoding established saved configuration state without running Packet Tracer or changing the originals. The file inspector verifies persisted state, not the timing of earlier experiments.

4.2 Testbed setup, configuration and parameters

Link / interface pair	Network	Endpoint addresses
R1 G0/0 – R2 G0/0	10.0.12.0/30	10.0.12.1 / 10.0.12.2
R2 G0/1 – R3 G0/0	10.0.23.0/30	10.0.23.1 / 10.0.23.2
R3 G0/1 – R4 G0/0	10.0.34.0/30	10.0.34.1 / 10.0.34.2
R4 G0/1 – R1 G0/1	10.0.41.0/30	10.0.41.1 / 10.0.41.2

LAN	Router G0/2 / gateway	PC address / mask
1	192.168.1.1	192.168.1.10 /24
2	192.168.2.1	192.168.2.10 /24
3	192.168.3.1	192.168.3.10 /24
4	192.168.4.1	192.168.4.10 /24

Transit masks are 255.255.255.252 and LAN masks are 255.255.255.0. The saved PC addresses and gateways match the written IP plan. The original base file has no dynamic routing process. All four routers in v2_rip.pkt contain RIP only, while all four in v3_ospf.pkt contain OSPF only. Running and startup configurations agree in both protocol files. The base file's running/startup contents differ, so it should be used as the preserved topology stage rather than assuming reload behaviour.

The actual OSPF commands use a broad transit match, network 10.0.0.0 0.255.255.255 area 0, plus the local LAN /24 wildcard. The guide instead lists separate /30 matches and explicit router IDs. No explicit router-id commands appear in the saved OSPF configurations. E27 shows neighbour IDs 192.168.2.1 and 192.168.4.1. These differences are recorded rather than silently substituting intended commands. Appendix A separates saved configuration excerpts from guide examples.

4.3 Code structure and explanation of key modules

Module	Role
Dashboard/index.html	Semantic sections, controls, figures, chart canvases and accessible dialog
Dashboard/css/style.css	Responsive layouts, protocol/status styling, focus states and reduced-motion rules
Dashboard/js/data.js	Local data constant mirroring CSV/JSON plus evidence and provenance
Dashboard/js/app.js	Scenario selection, illustrative SVG paths, evidence filters, lightbox and Chart.js views

Module	Role
Scripts/generate_charts.py	Deterministic static figures from canonical CSV, including hatched assumptions
Scripts/validate_data.py	Evidence/file hashes, mirrors, arithmetic and assumed-status invariants

The dashboard opens by double-clicking index.html. It uses no backend, runtime network request, fetch call or server. The measured-only toggle removes assumed cases and the mixed descriptive average. Experiment Player selections for assumed links show an explicit no-recorded-evidence card. E22 remains visible only in the evidence library with a needs-confirmation flag, and is not attached to restoration results.

4.4 Challenges and how they were handled

The guide’s claim that the ring removes equal-cost ambiguity does not hold for opposite LANs. E07 and E23 show R1’s equal-cost routes to LAN3 via R2 and R4. The recorded router-failure traffic is also PC1→PC3 with R4 off, rather than the guide’s PC2→PC4 example. The report preserves the performed experiment and treats the individual forwarding path as unverified.

PC ping syntax differs from IOS syntax. Captures show the PC accepting ping -t followed by the destination. Stopwatch timing is separate from simulator ping RTT, route-age fields and timeout counts. Individual interface costs are not displayed in the screenshots, so the cumulative OSPF costs are reported without asserting that the screenshots directly prove their component costs. E22’s startup timeout prevents its use as zero-loss restoration proof.

5 EXPERIMENTAL SETUP AND PERFORMANCE EVALUATION

5.1 Performance metrics and rationale

OCRT describes end-to-end service recovery as visible to continuous ping. The external stopwatch starts at failure introduction and stops at the first successful reply followed by at least one more consecutive reply. It includes detection, routing, forwarding and end-host effects. Packet behaviour is recorded qualitatively where excerpts show timeouts or unreachable responses. Loss counts and percentages require completed ping summaries, which are absent. RIP hop metric and OSPF cost are shown separately because their units differ.

Convergence values reported in this project are Observed Connectivity Recovery Times, measured via external stopwatch against Packet Tracer's simulated continuous ping — not laboratory-grade, isolated routing-protocol convergence measurements. They include human stopwatch reaction time (approx. 0.3–1 second), Packet Tracer's own simulation/event-processing delay, and end-to-end effects (link-down detection, routing recalculation, ARP refresh) rather than pure routing-protocol convergence alone. The metric remains valid for comparing RIP and OSPF because the identical method is applied to both, under identical topology and failure conditions.

The guide paragraph’s identical-conditions claim applies only where the conditions were actually matched. Exp B compares an earlier OSPF observation with a later RIP run and therefore cannot support direct protocol ranking. OSPF restoration’s TTL-change interval is a distinct path-return observation rather than outage OCRT.

5.2 Scenarios and parameter values

R1–R2 was timed in Packet Tracer; the values for the other three links are estimates assumed from the R1–R2 observations and the expected behaviour of the protocols, because OCRT differs from run to run and those links were not individually timed.

Scenario	Action / traffic	Recorded scope
A	R1 G0/0 shutdown, PC1→PC2	R1–R2 timing and route change, one run per protocol

Scenario	Action / traffic	Recorded scope
Illustrative link cases	R2–R3, R3–R4, R4–R1	Supplied estimates only; no separately timed experiment
B	R4 off, PC1→PC3	RIP later timing; OSPF earlier no-loss observation
C	R1–R2 re-enabled, PC1→PC2	Preferred-route return; OSPF TTL change; RIP timing unreliable

5.3 Results, tables and graphs

Table 5.1 adapts the template's baseline/proposed columns to RIP/OSPF and records provenance. "Single run" replaces the repeated-run mean ± SD expectation. The arithmetic average across supplied links appears separately and includes assumed estimates; it is not a trial average. No SD or error bars are calculated.

Scenario	Metric	RIP	OSPF	Observation	Data status
R1–R2	OCRT	≈26 s	≈7 s	73.08% shorter OSPF OCRT in these runs	Measured, single run
R2–R3	Supplied time	24 s	6 s	75.00% shorter, derived from estimates	Assumed (estimated – not measured)
R3–R4	Supplied time	28 s	8 s	71.43% shorter, derived from estimates	Assumed (estimated – not measured)
R4–R1	Supplied time	25 s	7 s	72.00% shorter, derived from estimates	Assumed (estimated – not measured)

Table 5.1. Supplied link-failure timing comparison. Assumed (estimated – not measured) applies to every numeric input and derived comparison in the three estimated-link rows.

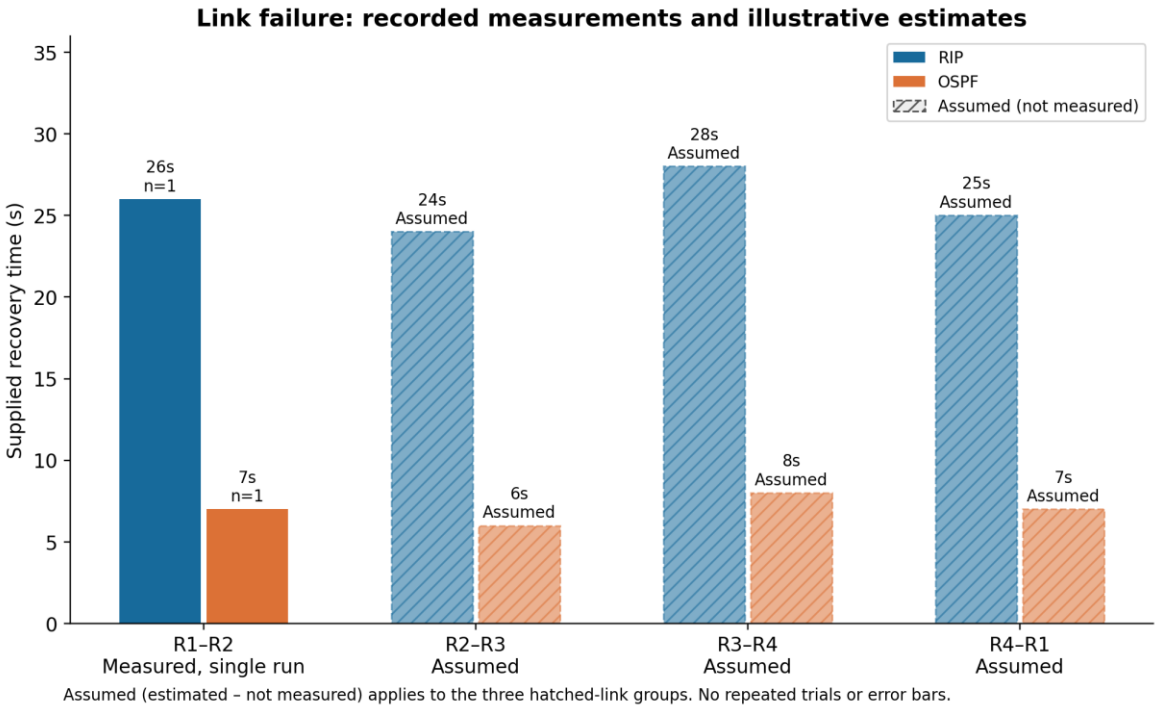


Fig. 5.1. Link timings. Solid = recorded, single run; hatched = Assumed (estimated – not measured).

Summary	RIP	OSPF	Interpretation
Arithmetic average across four links	25.75 s	7.00 s	Includes three Assumed (estimated – not measured) link inputs
Reduction of averages / ratio	18.75 s gap	72.82% / 3.68×	Illustrative descriptive arithmetic, three Assumed inputs; no repeated-trial or statistical claim

The average is (26 + 24 + 28 + 25)/4 for RIP and (7 + 6 + 8 + 7)/4 for OSPF. In both sums, the final three terms are Assumed (estimated – not measured). The 72.82% reduction uses the two averages, rather than averaging per-link percentages. Every derived result inherits its input status.

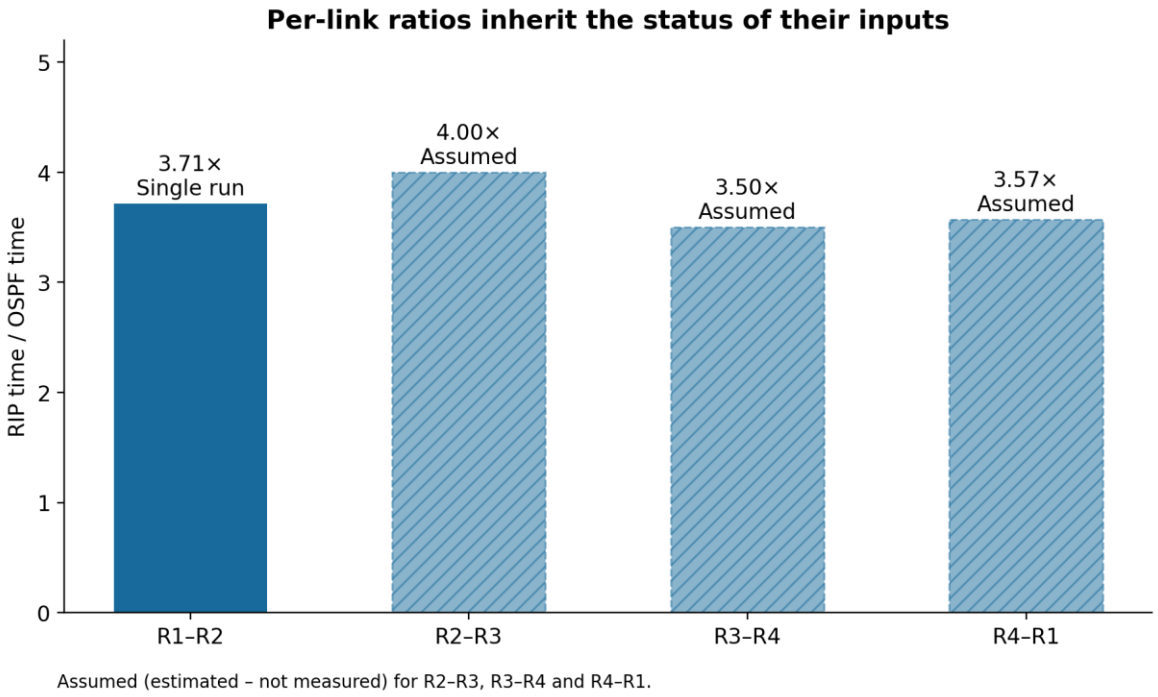


Fig. 5.2. Per-link RIP/OSPF ratios. R1–R2 is single run; other ratios use Assumed (estimated – not measured) inputs.

Figures 5.3 and 5.4 establish the before/after route change for the measured link. The stopwatch durations come from the recorded results tables; route-age fields in these screenshots are not used as timers.

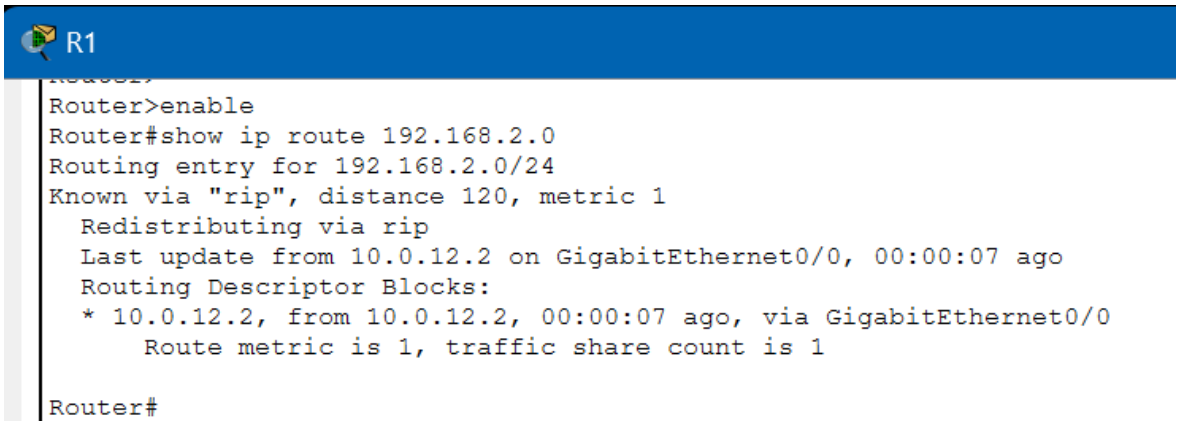


Fig. 5.3. RIP before R1–R2 failure: next hop 10.0.12.2, metric 1. Recorded evidence (Packet Tracer), E11.

```
Router#enable
Router#show ip route 192.168.2.0
Routing entry for 192.168.2.0/24
Known via "rip", distance 120, metric 3
  Redistributing via rip
  Last update from 10.0.41.1 on GigabitEthernet0/1, 00:00:02 ago
  Routing Descriptor Blocks:
  * 10.0.41.1, from 10.0.41.1, 00:00:02 ago, via GigabitEthernet0/1
      Route metric is 3, traffic share count is 1
```

Fig. 5.4. RIP alternate route: next hop 10.0.41.1, metric 3. Recorded evidence (Packet Tracer), E14.

```
Router>enable
Router#show ip route 192.168.2.0
Routing entry for 192.168.2.0/24
Known via "ospf 1", distance 110, metric 2, type intra area
  Last update from 10.0.12.2 on GigabitEthernet0/0, 00:35:14 ago
  Routing Descriptor Blocks:
  * 10.0.12.2, from 192.168.2.1, 00:35:14 ago, via GigabitEthernet0/0
      Route metric is 2, traffic share count is 1
```

Fig. 5.5. OSPF before failure: next hop 10.0.12.2, cost 2. Recorded evidence (Packet Tracer), E28.

```
Router#show ip route 192.168.2.0
Routing entry for 192.168.2.0/24
Known via "ospf 1", distance 110, metric 4, type intra area
  Last update from 10.0.41.1 on GigabitEthernet0/1, 00:02:31 ago
  Routing Descriptor Blocks:
  * 10.0.41.1, from 192.168.2.1, 00:02:31 ago, via GigabitEthernet0/1
      Route metric is 4, traffic share count is 1

Router#
```

Fig. 5.6. OSPF alternate route: next hop 10.0.41.1, cost 4. Recorded evidence (Packet Tracer), E31.

Scenario	Metric	RIP	OSPF	Observation	Data status
R4 off	OCRT	≈11 s, later run	No loss observed; OCRT not measurable, earlier run	Unlike runs; do not rank protocols	Recorded observation; RIP timed once
R1–R2 restored	Path return	Original route; time not reliable	≈11 s to TTL 124→126	Path-change interval, not outage OCRT	Recorded route behaviour; OSPF single interval

Table 5.2. Router failure and restoration with non-comparability and missing-timing disclosures. No numeric zero substitutes for OSPF's no-loss observation or RIP's unreliable restoration interval.

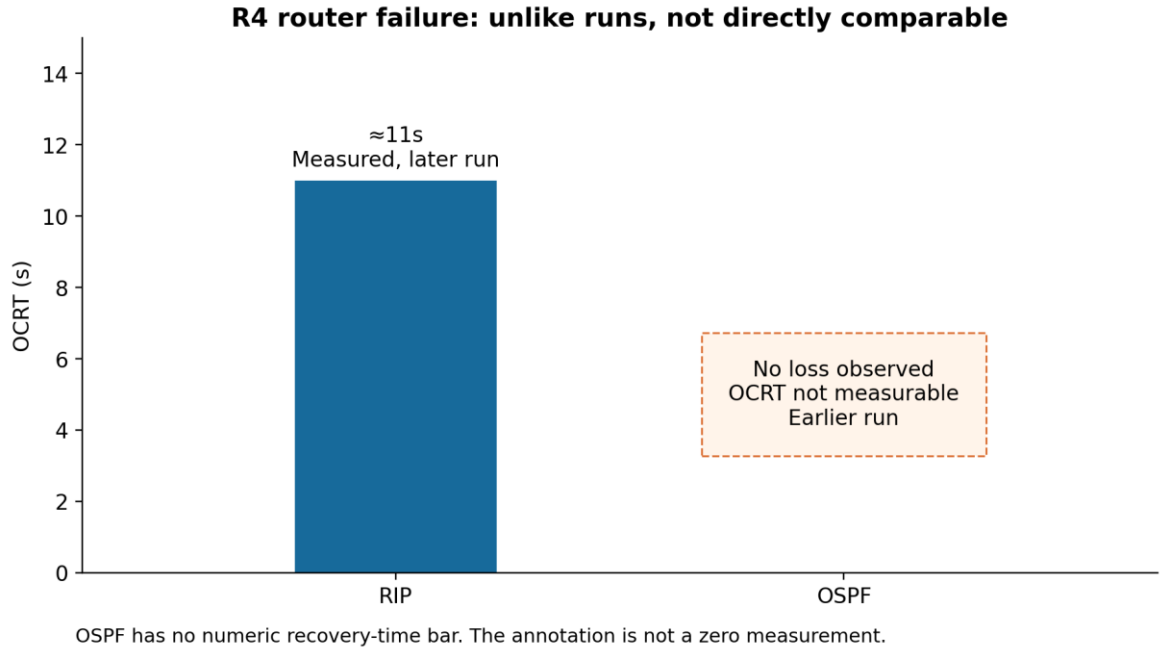


Fig. 5.7. R4 failure: RIP later run and OSPF earlier run are not directly comparable. No OSPF zero bar.

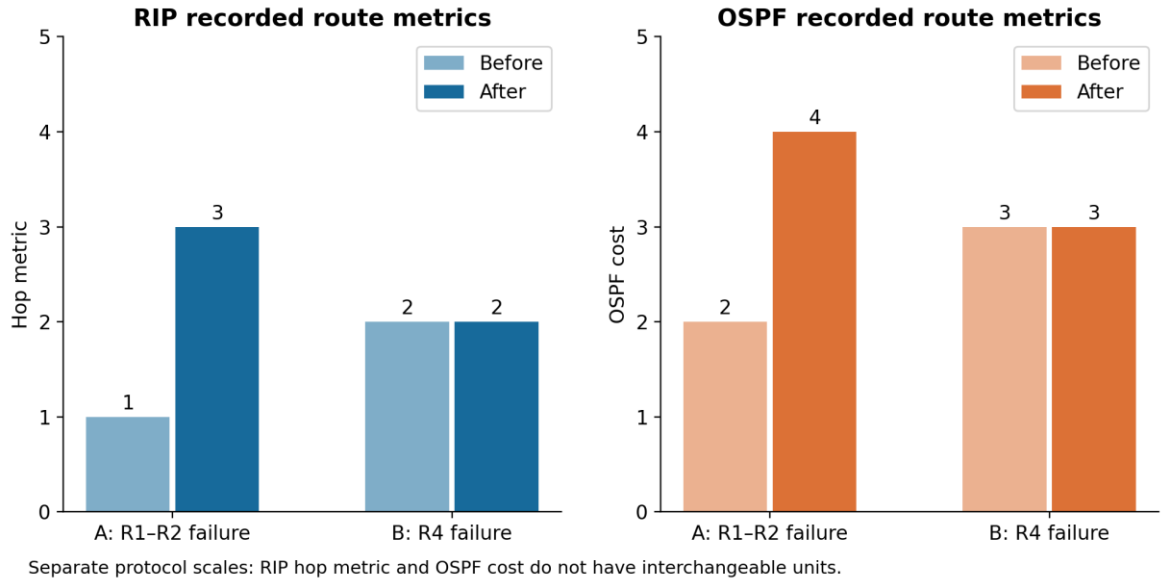


Fig. 5.8. Recorded before/after metrics on separate RIP hop-metric and OSPF-cost panels.

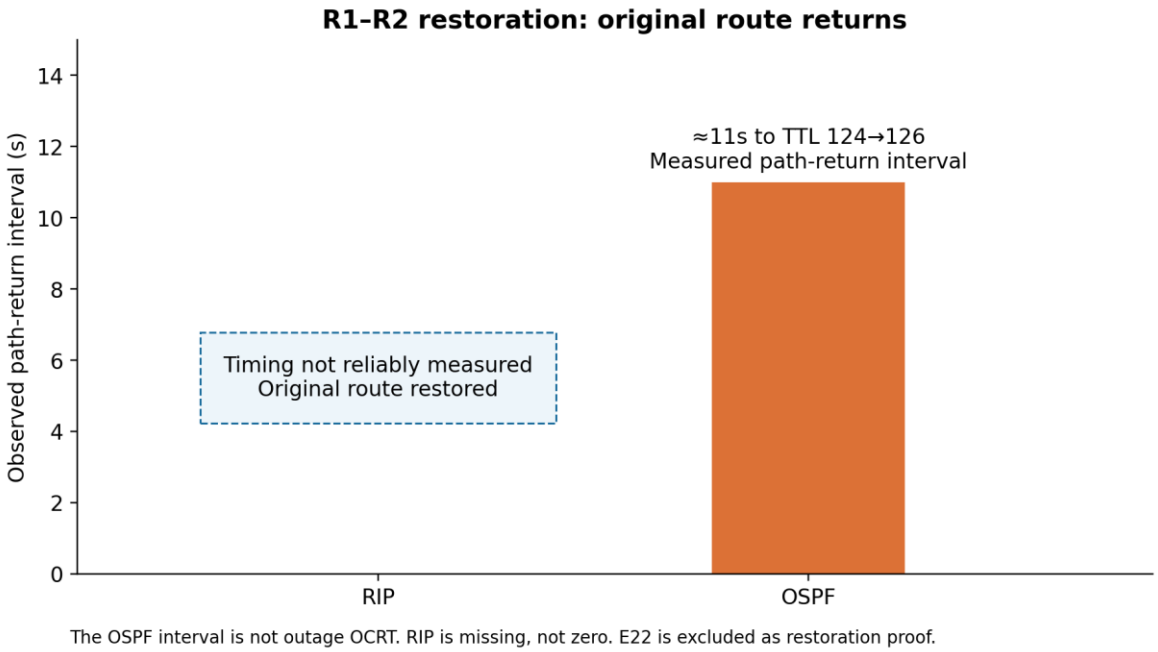


Fig. 5.9. Restored route: OSPF interval to TTL/path change; RIP timing not reliably measured.

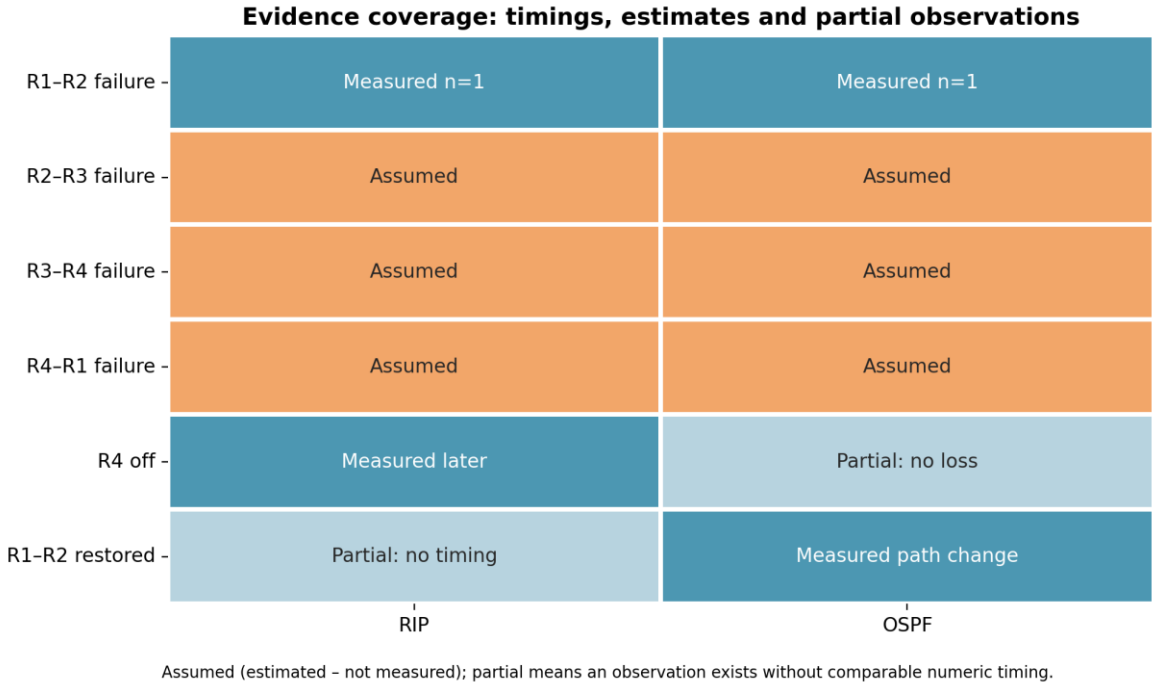


Fig. 5.10. Evidence coverage. Assumed (estimated – not measured) cases have no recorded timing evidence.

5.4 Analysis and discussion

For the recorded R1–R2 failure, OSPF's approximately 7-second OCRT is shorter than RIP's approximately 26-second OCRT. The per-run reduction is (26−7)/26 ×100 =73.08%, with no

claim of statistical significance. RIP's captured output contains unreachable responses and timeouts; OSPF's captured excerpt has one timeout before replies continue. Neither excerpt supplies a completed loss summary.

RIP's periodic/triggered distance-vector updates and OSPF's triggered link-state processing offer a theoretical explanation consistent with the observation [1], [2], [4], [5]. The shutdown captures explicitly show OSPF reacting to interface-down, so it would be incorrect to assume that a full Dead timer necessarily expired. The experiment does not isolate the contributions of protocol processing, ARP, detection, simulator scheduling or the stopwatch operator.

All supplied link values favour OSPF numerically, but the three estimated cases cannot independently corroborate the measured comparison. Their averages and ratios describe the team's illustrative assumptions. For router failure, equal-cost paths before R4 failure mean traffic could already have used R2. That is a plausible but unverified hypothesis for the earlier OSPF no-loss observation, not a proven explanation.

5.5 Critical evaluation and data authenticity

R1–R2 was timed in Packet Tracer; the values for the other three links are estimates assumed from the R1–R2 observations and the expected behaviour of the protocols, because OCRT differs from run to run and those links were not individually timed.

The strengths are a common verified topology, separate saved protocol configurations, visible before/after route output, preserved original images, and an offline replay linked to those sources. The main weaknesses are one timed run for each reported measurement, three assumed-link estimates, unmatched router-failure runs, missing loss summaries and the simulator/stopwatch measurement boundary. No standard deviation or confidence interval is justified by the dataset.

Statement on data authenticity and limitations: captured screenshots establish visible routing and ping behaviour; results tables supply the reported stopwatch values. Assumed (estimated – not measured) values are illustrative estimates and never carry experiment evidence IDs. E22 remains needs-confirmation and is excluded as restoration proof. Numeric gaps remain non-numeric. The mixed four-link arithmetic average was requested as a descriptive summary and cannot establish repeatability or general superiority.

6 CONCLUSION AND FUTURE WORK

6.1 Summary of work and key findings

The preserved Packet Tracer files verify a common ring with a RIP-only RIPv2 configuration and an OSPF-only process 1 / area 0 configuration. In the measured R1–R2 single runs, OCRT was approximately 26 seconds for RIP and 7 seconds for OSPF. Both protocols switched to the alternate ring route and later restored the direct route. Across four supplied links, including three Assumed (estimated – not measured) links, descriptive averages are 25.75 and 7.00 seconds, with a 72.82% reduction of averages. Those arithmetic summaries are illustrative rather than repeated-run experimental estimates.

6.2 Limitations

Single runs do not characterise variation. The other three link cases are assumed, loss percentages are missing, and the router-failure timings come from unlike runs. Stopwatch reaction and Packet Tracer event processing are inseparable from observed service recovery. OSPF's restoration timing refers to TTL/path change. Individual active forwarding decisions were not recorded, and no event-list capture proves a particular update-timer mechanism caused the timing difference.

6.3 Future enhancements

Time every ring link independently and repeat each matched scenario at least ten times before reporting repeated-trial mean and SD. Capture the exact failure trigger, full ping summaries and automated timestamps. Verify the active forwarding path before failing a router. Subsequent work can compare tuned timers, BFD, EIGRP or multi-area OSPF and validate the observations with emulated or real routers. Both team members should repeat the complete explanation across protocols, code and limitations.

7 REFERENCES

[1] G. Malkin, “RIP Version 2,” RFC 2453, Nov. 1998, doi:10.17487/RFC2453. Available: https://www.rfc-editor.org/info/rfc2453/

[2] J. Moy, “OSPF Version 2,” RFC 2328, Apr. 1998, doi:10.17487/RFC2328. Available: https://www.rfc-editor.org/info/rfc2328/

[3] C. Hedrick, “Routing Information Protocol,” RFC 1058, Jun. 1988, doi:10.17487/RFC1058. Available: https://www.rfc-editor.org/info/rfc1058/

[4] Cisco, “Cisco IOS IP Routing: RIP Command Reference: RIP Commands,” online documentation, accessed Oct.26,2026. Available: https://www.cisco.com/c/en/us/td/docs/ios/iproute_rip/command/reference/irr_book/irr_rip.html

[5] Cisco, “IP Routing: OSPF Configuration Guide, Cisco IOS XE Release 2,” online documentation, accessed Oct.26,2026. Available: https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/iproute_ospf/configuration/xe-2/iro-xe-2-book.pdf

APPENDIX A: CONFIGURATION LISTINGS

A.1 Verified saved-file excerpts. These lines were extracted read-only from the RUNNINGCONFIG elements of v2_rip.pkt and v3_ospf.pkt. They are not reconstructed from the guide or presented as configuration screenshots. Full saved running-config text for every router is provided in Configs/.

RIP routing sections for R1–R4

R1 — RIP saved configuration

```
router rip
 version 2
 network 10.0.0.0
 network 192.168.1.0
 no auto-summary
```

R2 — RIP saved configuration

```
router rip
 version 2
 network 10.0.0.0
 network 192.168.2.0
 no auto-summary
```

R3 — RIP saved configuration

```
router rip
 version 2
 network 10.0.0.0
 network 192.168.3.0
 no auto-summary
```

R4 — RIP saved configuration

```
router rip
 version 2
 network 10.0.0.0
 network 192.168.4.0
 no auto-summary
```

OSPF routing sections for R1–R4

R1 — OSPF saved configuration

```
router ospf 1
 log-adjacency-changes
 network 10.0.0.0 0.255.255.255 area 0
 network 192.168.1.0 0.0.0.255 area 0
```

R2 — OSPF saved configuration

```
router ospf 1
 log-adjacency-changes
 network 10.0.0.0 0.255.255.255 area 0
 network 192.168.2.0 0.0.0.255 area 0
```

R3 — OSPF saved configuration

```
router ospf 1
 log-adjacency-changes
 network 10.0.0.0 0.255.255.255 area 0
 network 192.168.3.0 0.0.0.255 area 0
```

R4 — OSPF saved configuration

```
router ospf 1
 log-adjacency-changes
 network 10.0.0.0 0.255.255.255 area 0
 network 192.168.4.0 0.0.0.255 area 0
```

All router display names are R1–R4, while the saved IOS hostname is Router. No explicit router-id command appears in the saved OSPF configuration. The guide's intended explicit router IDs were not substituted into these verified excerpts.

A.2 Intended guide examples, not verified entered commands

```
! Guide-intended R1 example; not a transcript
router ospf 1
 router-id 1.1.1.1
 network 10.0.12.0 0.0.0.3 area 0
 network 10.0.41.0 0.0.0.3 area 0
 network 192.168.1.0 0.0.0.255 area 0
```

A.3 Captured CLI actions. E12/E29 show interface GigabitEthernet0/0 followed by shutdown; E16/E33 show no shutdown. Baseline screenshots show show ip route and E27 shows show ip ospf neighbor. The screenshot set does not contain full running-config commands. Saved-file extraction, rather than screenshot reconstruction, supplies the complete configuration listings.

APPENDIX B: EVIDENCE AND RAW RESULTS

B.1 Additional recorded screenshots. All 37 lossless originals are supplied in Dashboard/assets/screenshots. The manifest below maps every source image. Only selected readable figures are embedded in this report; none is used as evidence for an assumed timing.

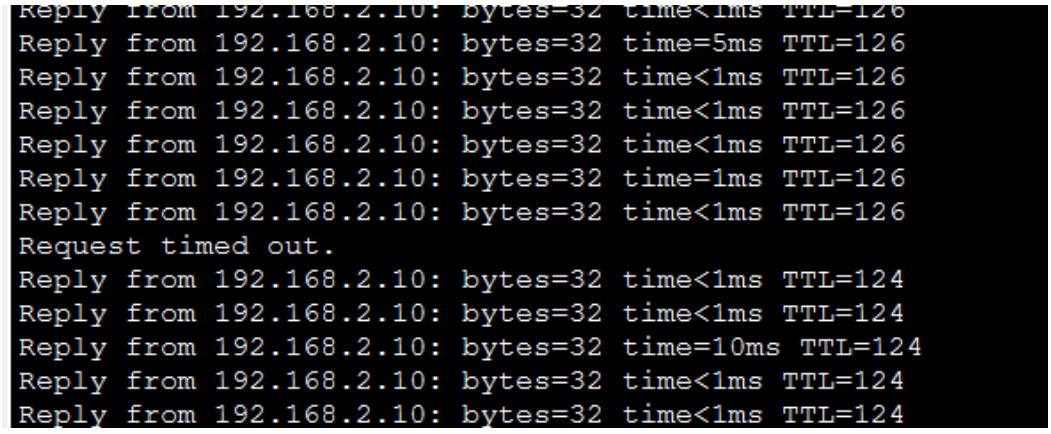


Fig. B.1. OSPF link-failure ping excerpt with one visible timeout and TTL126→124. Recorded evidence (Packet Tracer), E30.

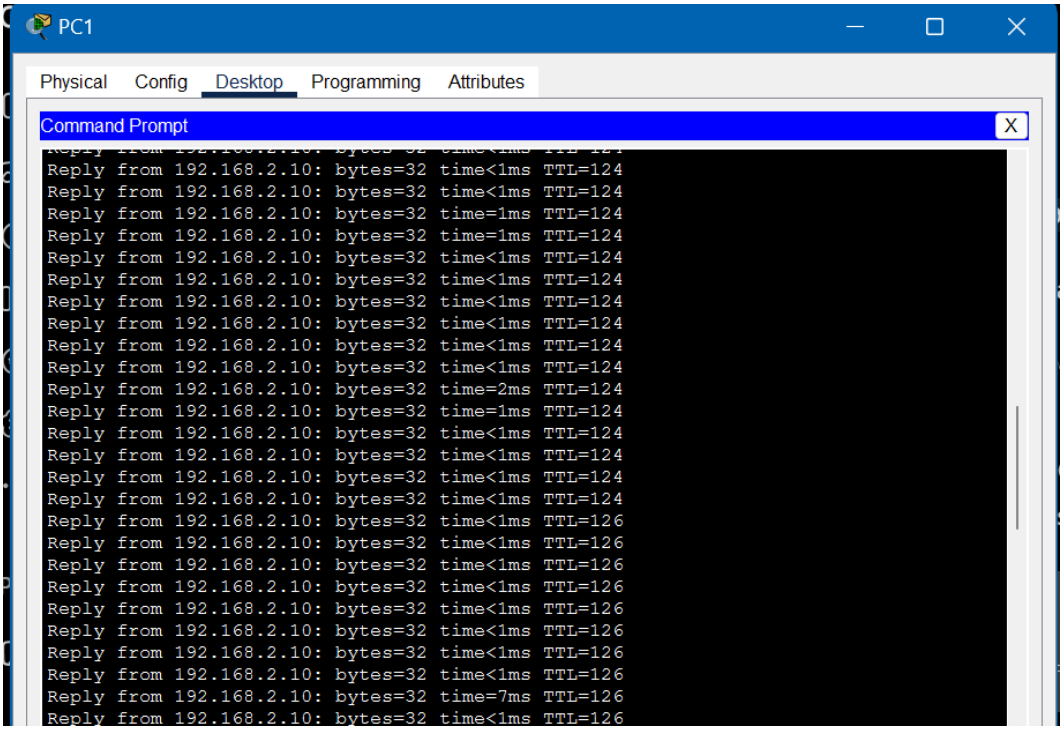


Fig. B.2. OSPF restoration TTL124→126; reported ≈11s interval is supplied by the results table. Recorded evidence (Packet Tracer), E04.

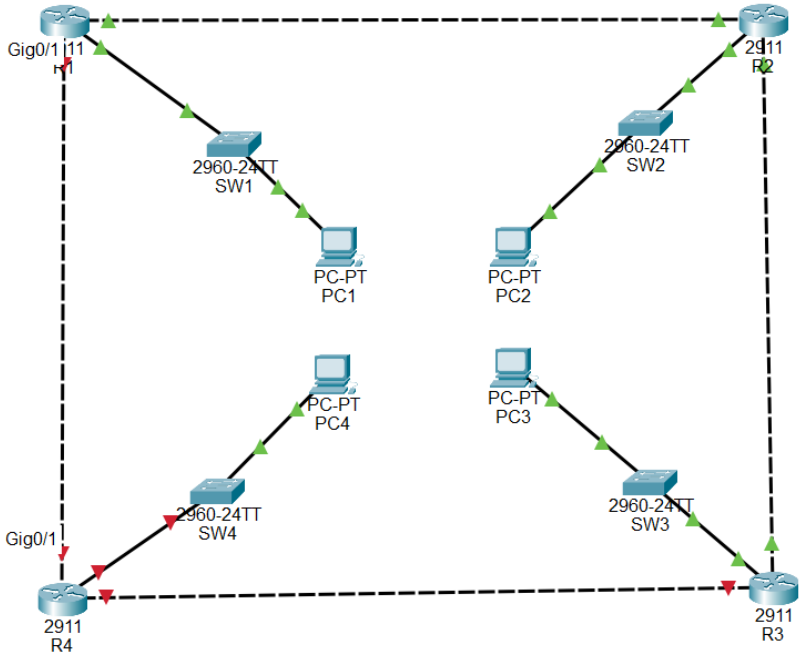


Fig. B.3. R4 unavailable, with its incident links down in the RIP test context. Recorded evidence (Packet Tracer), E19.

```
Router>enable
Router#show ip route 192.168.3.0
Routing entry for 192.168.3.0/24
Known via "ospf 1", distance 110, metric 3, type intra area
  Last update from 10.0.12.2 on GigabitEthernet0/0, 00:03:05 ago
  Routing Descriptor Blocks:
  * 10.0.12.2, from 192.168.3.1, 00:03:05 ago, via GigabitEthernet0/0
      Route metric is 3, traffic share count is 1

Router#
```

Fig. B.4. OSPF surviving route to LAN3 via R2 with cost3. Recorded evidence (Packet Tracer), E37.

B.2 Raw supplied timing rows

Case	Protocol	Value / status	Data status	Evidence IDs
A R1–R2	RIP	26 s	Recorded single run	E11;E12;E13;E14;E15
A R1–R2	OSPF	7 s	Recorded single run	E28;E29;E30;E31;E32

Case	Protocol	Value / status	Data status	Evidence IDs
A2 R2–R3	RIP	24 s	Assumed (estimated – not measured)	None
A2 R2–R3	OSPF	6 s	Assumed (estimated – not measured)	None
A3 R3–R4	RIP	28 s	Assumed (estimated – not measured)	None
A3 R3–R4	OSPF	8 s	Assumed (estimated – not measured)	None
A4 R4–R1	RIP	25 s	Assumed (estimated – not measured)	None
A4 R4–R1	OSPF	7 s	Assumed (estimated – not measured)	None
B R4	RIP	11 s	Recorded single run	E07;E18;E19;E20;E21
C R1–R2	RIP	Not reliably measured	Recorded qualitative observation; timing absent	E14;E16;E17
B R4	OSPF	Not measurable	Recorded qualitative observation; timing absent	E23;E34;E35;E36;E37
C R1–R2	OSPF	11 s	Recorded single run	E01;E02;E03;E04;E05;E33

For OSPF C the value is a path-return/TTL interval. For B, RIP is the later run and OSPF the earlier run. The complete CSV retains routes, metrics, run notes, counts and evidence fields. No packet-loss count/percentage is recorded in a completed screenshot summary.

B.3 Evidence manifest

ID	Protocol / case / phase	Caption / confidence
E01	OSPF / restoration / before	Initial path; confirmed
E02	OSPF / restoration / during	Putting the r1->r2 link down; confirmed
E03	OSPF / restoration / after link failure	New path; confirmed
E04	OSPF / restoration / during/after	TTL change; confirmed
E05	OSPF / restoration / after	Confirming the initial path is taken again; confirmed
E06	Topology / baseline / proof	Topology and IP addressing; confirmed
E07	RIP / baseline / before	R1 RIP; confirmed
E08	RIP / baseline / before	R2 RIP; confirmed
E09	RIP / baseline / before	R3 RIP; confirmed
E10	RIP / baseline / before	R4 RIP; confirmed
E11	RIP / link failure / before	RIP route before failure; confirmed
E12	RIP / link failure / during	Disabling the r1->r2 link; confirmed
E13	RIP / link failure / during/after	Disabling the r1->r2 link; confirmed
E14	RIP / link failure / after	New path after r1->r2 was disabled; confirmed
E15	RIP / link failure / proof	Visual proof that r1->r2 was disabled; confirmed
E16	RIP / restoration / during	Restoring the r1->r2 link; confirmed
E17	RIP / restoration / after	Verifying that its back; confirmed
E18	RIP / router failure / during/after	RIP router failure test; confirmed
E19	RIP / router failure / proof	Visual evidence that r4 is down; confirmed

ID	Protocol / case / phase	Caption / confidence
E20	RIP / router failure / after	Verifying the change; confirmed
E21	RIP / router failure / proof	R4 is down; confirmed
E22	RIP / restoration / uncertain	Restoring ping for RIP; needs-confirmation
E23	OSPF / baseline / before	OSPF R1 baseline; confirmed
E24	OSPF / baseline / before	R2 baseline; confirmed
E25	OSPF / baseline / before	R3 baseline; confirmed
E26	OSPF / baseline / before	R4 baseline; confirmed
E27	OSPF / baseline / before	OSPF baseline neighbors; confirmed
E28	OSPF / link failure / before	OSPF link route before ; confirmed
E29	OSPF / link failure / during	Disabling the link; confirmed
E30	OSPF / link failure / during/after	Pinglog of PC1; confirmed
E31	OSPF / link failure / after	New route; confirmed
E32	OSPF / link failure / proof	Proof that the link is down; confirmed
E33	OSPF / restoration / during	Reenabling the link; confirmed
E34	OSPF / router failure / during/after	Pinglog for router failure testing; confirmed
E35	OSPF / router failure / proof	Proof that R4 is down for this test; confirmed
E36	OSPF / router failure / proof	Proof that R4 is down for this test; confirmed
E37	OSPF / router failure / after	Route after R4 is down; confirmed

The companion Data/evidence_manifest.csv provides original document indices, final filenames, SHA-256 hashes and human-reviewed what_it_shows descriptions for every image. Protocol assignment for a topology or PC crop can come from caption context. E22 is preserved and flagged, with no restoration result linked to it.